A context-sensitive deep learning approach for microcalcification detection in mammograms

INFO-H501 Pattern Recognition and Image Analysis

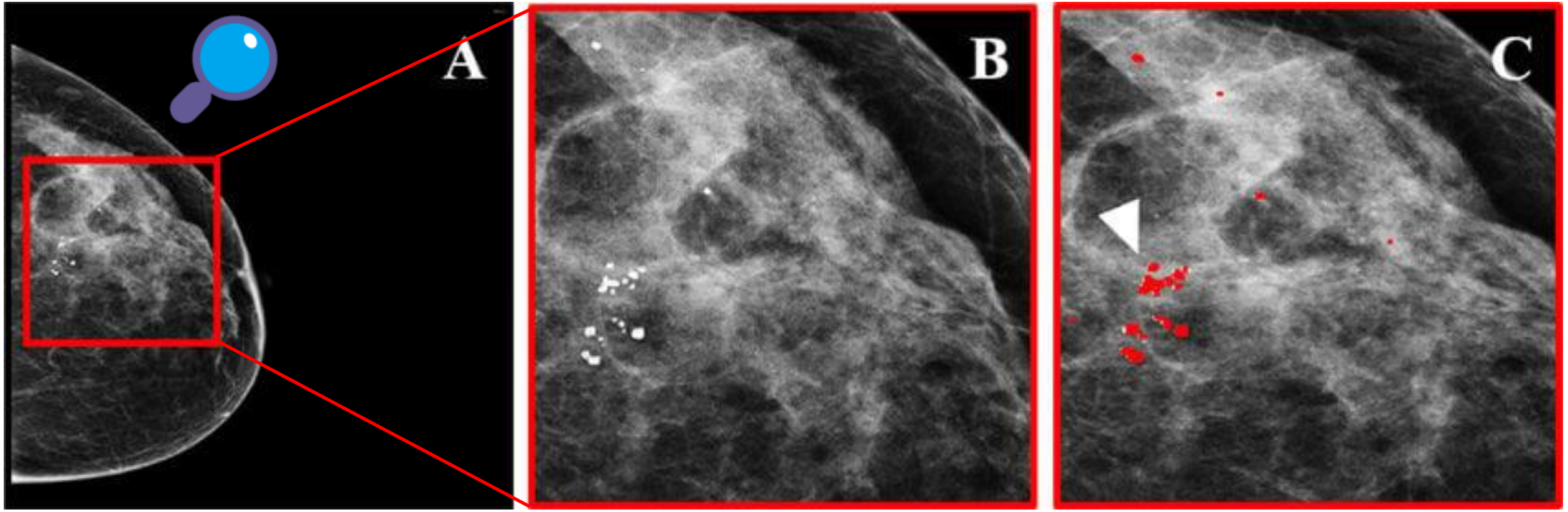
Theo Abraham
Kevin Straatman
Arthur Metzger
Bordis Paolo Alberto

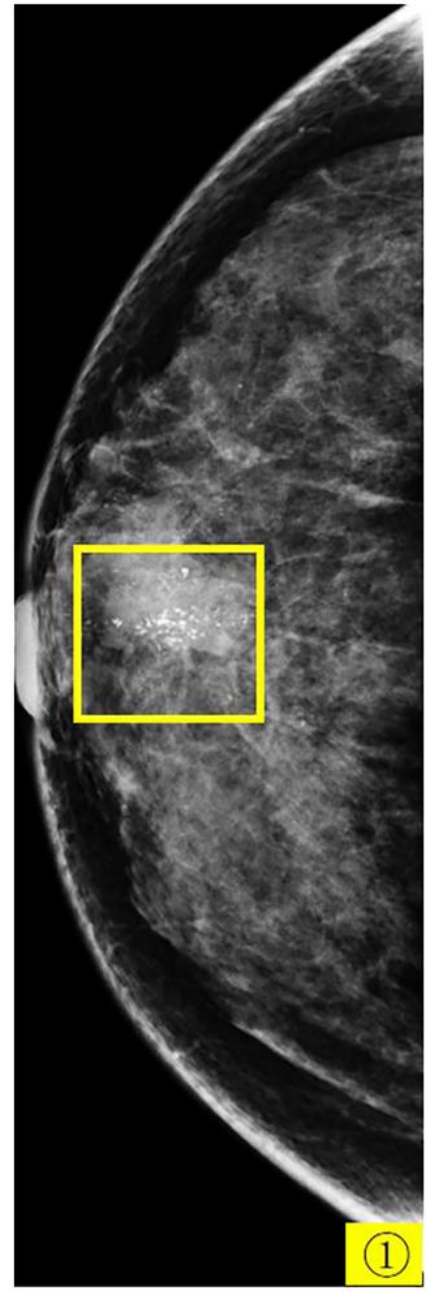


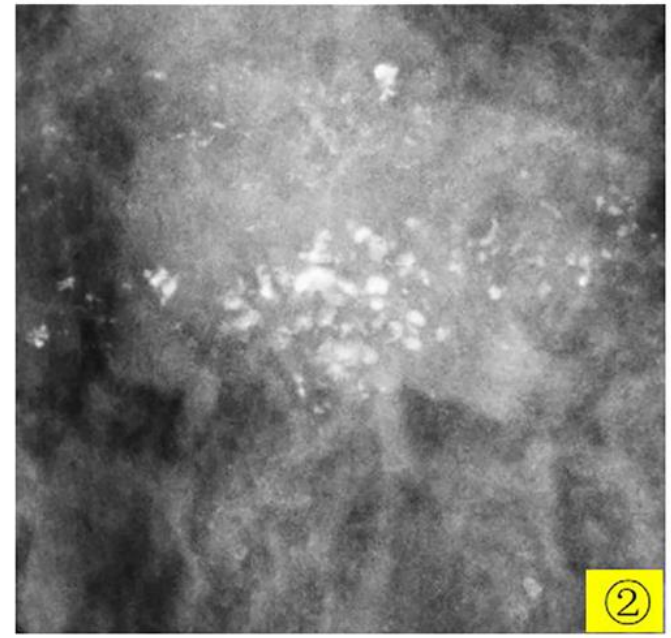
Note to us

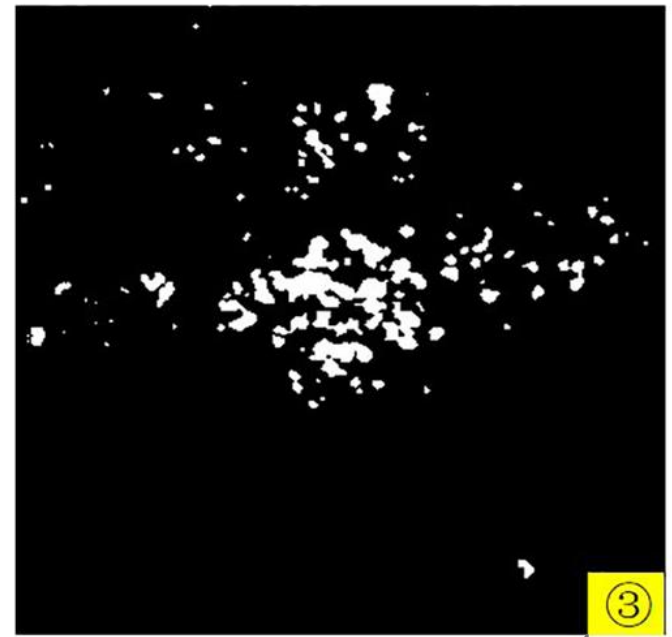
- Check out the notes for each slide
- Add small intro on MC ?

Background : Introduction to MC and MC Clusters




①


②


③

Background : The False Positives problem

- The False Positives (FPs) problem

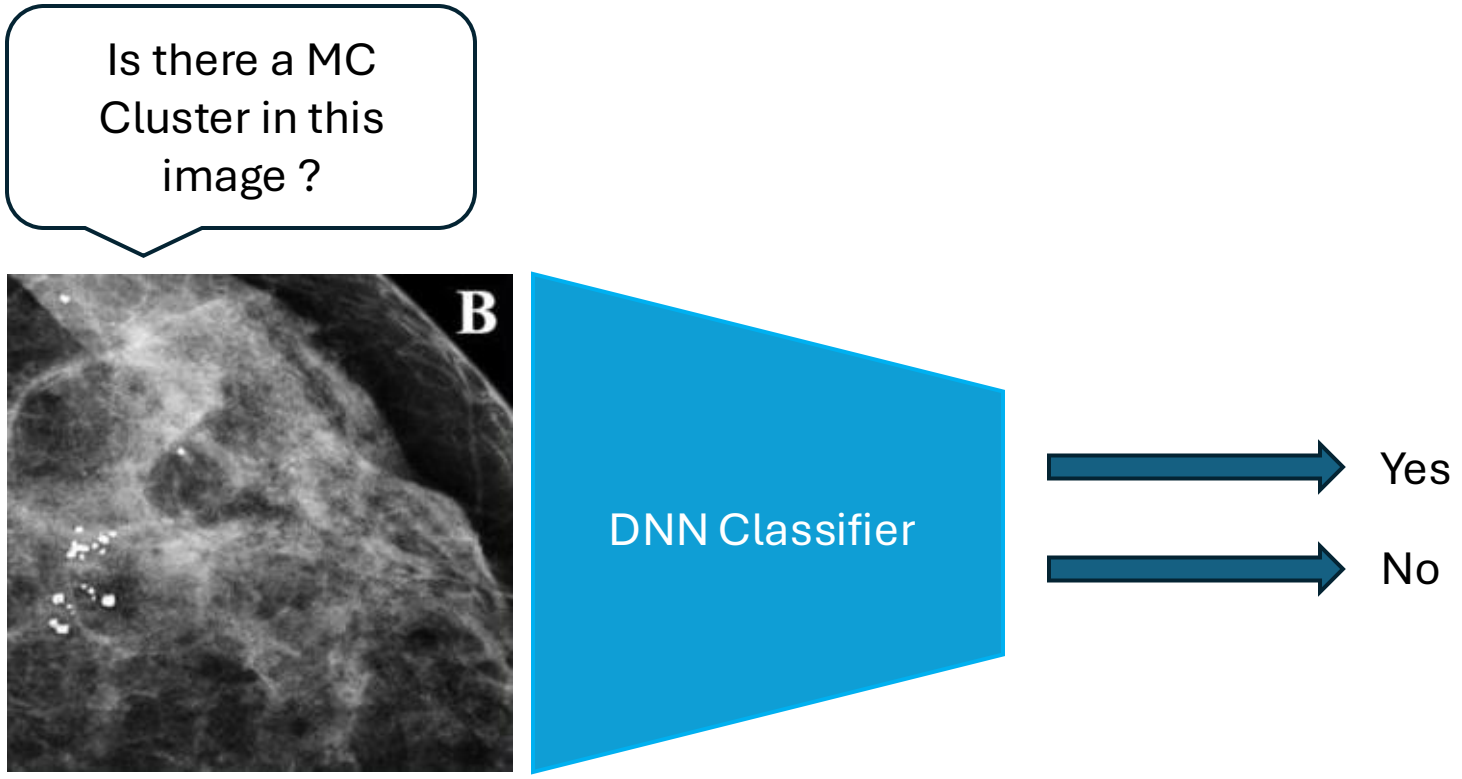
??? freeresponse receiver operating characteristic

1. *Unified SVM detector:* Recently developed in [9], this MC detector was designed to suppress occurrence of FPs caused by linear structures and MC-like noise patterns. It was demonstrated to yield improved performance over several existing MC detectors [9].

2. *CNN cluster detector:* This detector was developed for direct detection of the presence of clustered MCs in an image region based on deep learning [18]. It was demonstrated to outperform several existing MC based detectors in detecting MC clusters.

3. *A local DNN detector:* In contrast to the proposed context-sensitive DNN classifier, we also considered a DNN detector by using only the local subnetwork in the DNN architecture (Figure 3). This is to demonstrate the benefit of the additional use of the global subnetwork for context learning in MC detection. For convenience, this detector is referred to as local DNN. In the experiment, the local DNN classifier was trained with the same set of training samples by using the same procedure as in the context-sensitive DNN.

Background : previous CNN Classifier



Experiment – Dataset

- **Data Split**: 521 SFM and 188 FFDM images partitioned by case to avoid bias
  - Training: 167 cases (300 images)
  - Validation: 67 cases (117 images)
  - Test: 158 cases (292 images)

- **Data augmentation**: To improve DNN robustness, training samples were increased five-fold
  - Symmetry: Left-to-right and up-down flipping
  - Rotations: 90, 180, 270 rotations

- **Sample control**: MC samples were capped at 150 per mammogram to mantain class balance

Experiment

The Context-Sensitive approach relies on dual image windows co-centered at the detection location

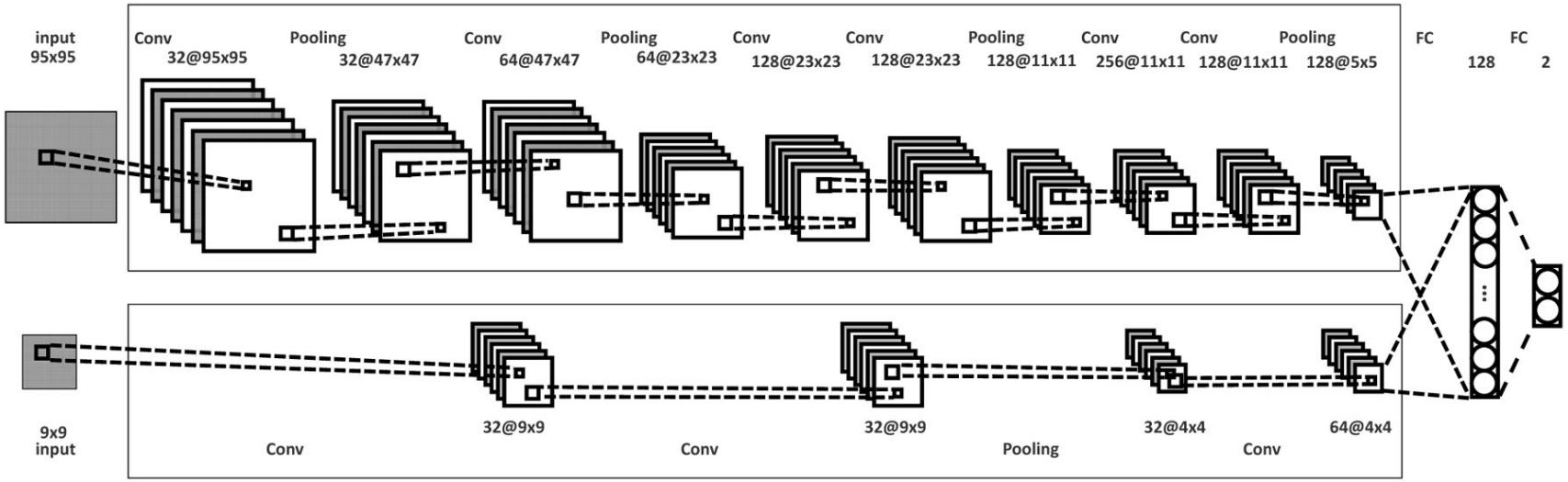
- **Local Window** :
  - Size: 9 x 9 pixels
  - Purpose: Captures the morphology of a single MC object
- **Global window:**
  - Size: 95 x 95 pixels (approx. 1cm^2)
  - Purpose: Characterizes the surrounding tissue background to suppress false positives (FPs)
- **Integration**: Features from both subnetworks are merged into fully-connected layers for final classification

Experiment – Test case

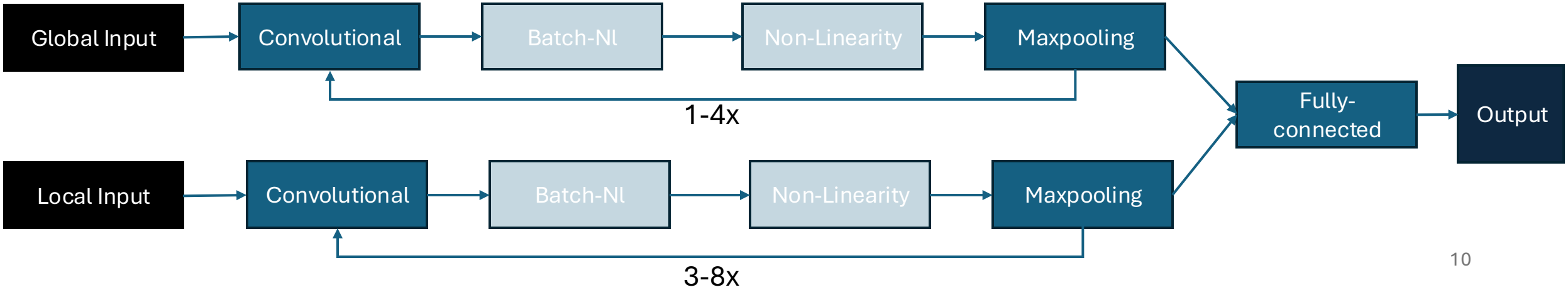
- **Test 1:** Single MC Detection (CADx Focus)
  - **Goal:** Accuracy of detecting individual MCs in a cluster
  - **Metric:** MC-based froc analysis
  - **Finding:** Achieved 81.9% reduction in FPs at 80% sensitivity compared to Unified SVM

- **Test 2:** MC Cluster Detection (CADe Focus)
  - **Goal:** Identifying suspicius cluster regions in full images
  - **Criteria:** Detections grouped with a 0.5cm connecting distance
  - **Metric:** Cluster-based FROC analysis

Model Architecture



Detailed layers :



Learning

Complete Dataset : MC & Non-MC images

Training Data | Test Data

Training set | Validation set

Model architecture

Model Training (Adam)

1 batch

n=256 training samples

Refined model params

1000x iteration
(Best model params)

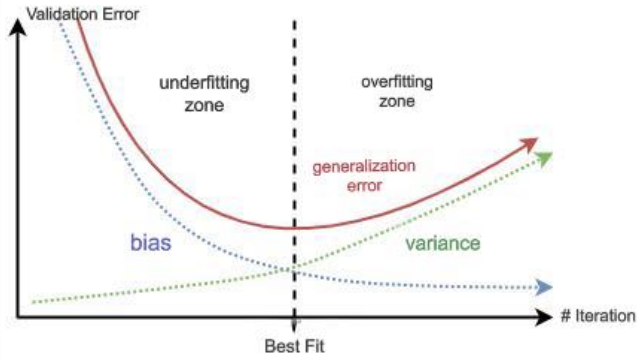
Model Validation

Validation set

Refined model hyperparams

Validation Error

20x iterations
(Best model Hyperparams)



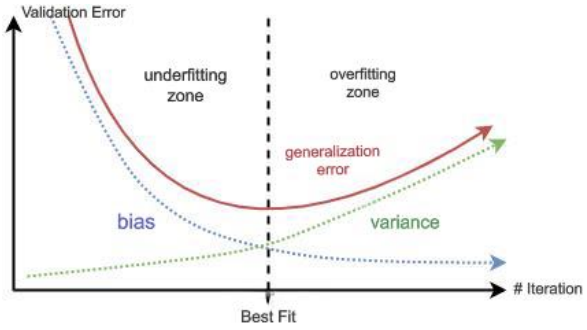
Final Test

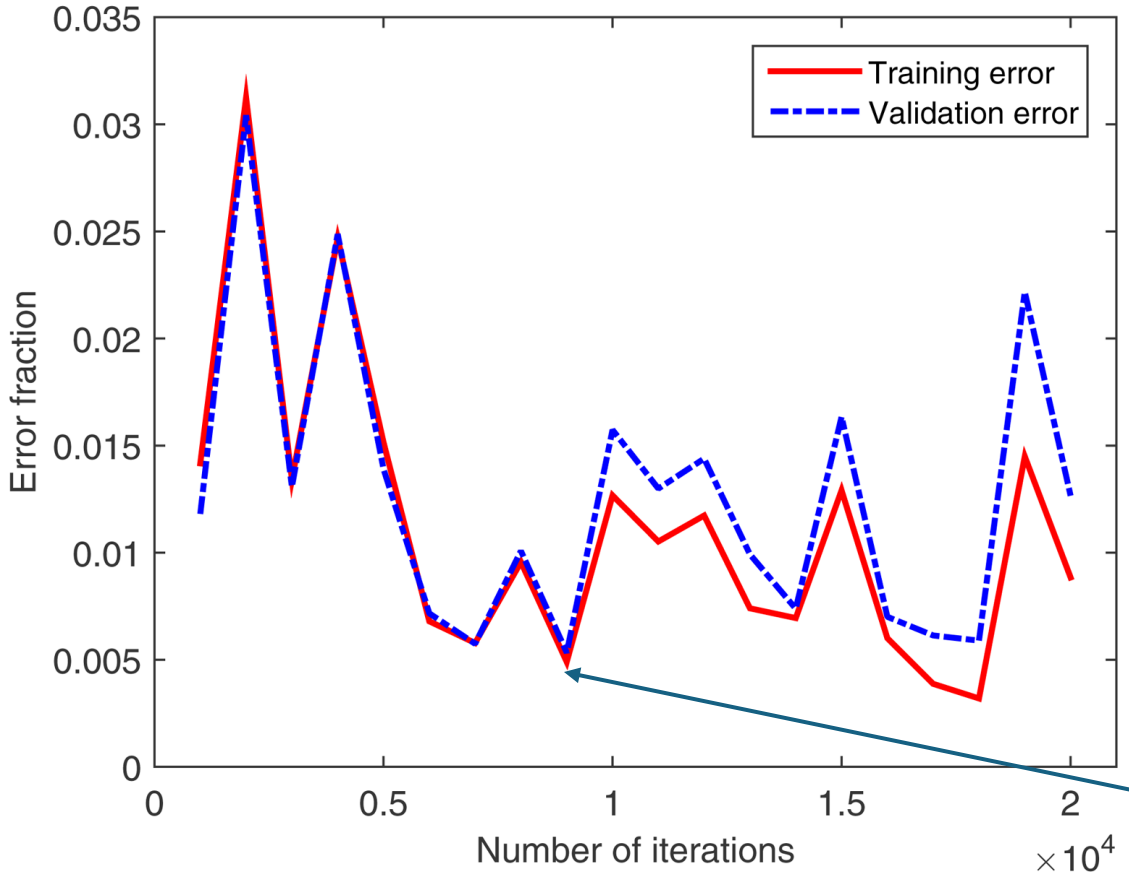
(Final Real-World performance of the Best Model architecture on unseen Data)

24 iterations*
(Best model architecture)

*Local : 4 layers (1-4)
Global : 6 layers (3-8

Validation Error plot



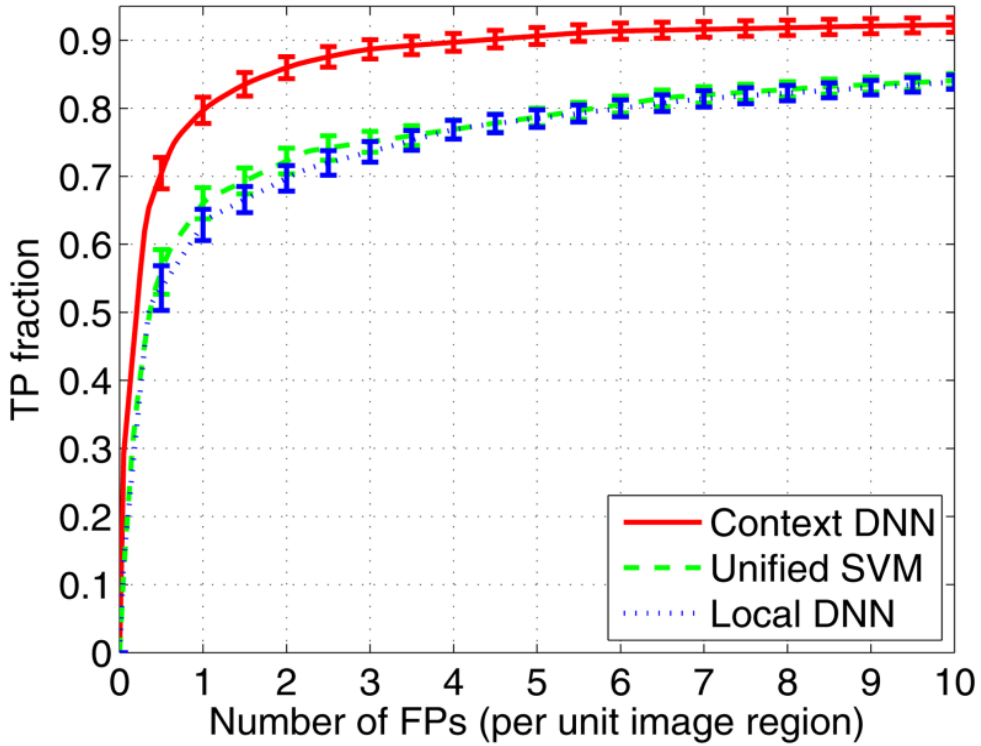


20 x 1000 steps

Lowest Validation error !

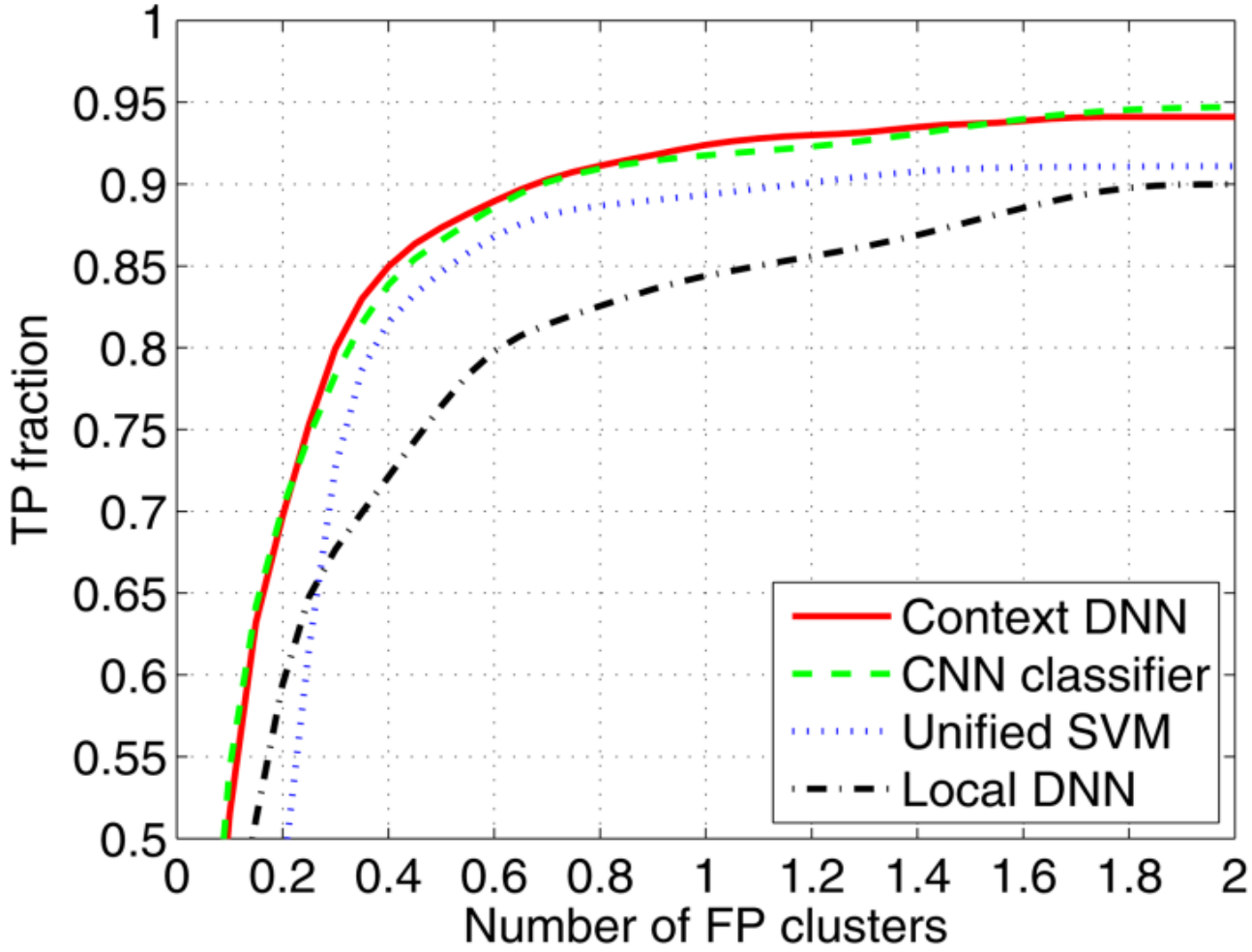
Results

individual

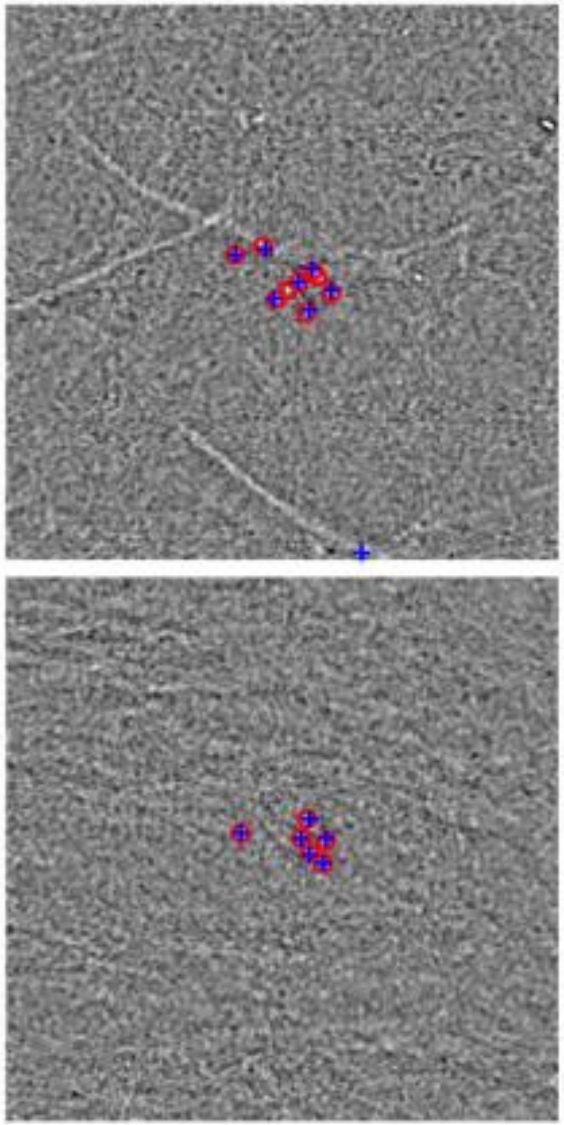


	Global subnetwork					
Local subnetwork	**3 Convs**	**4 Convs**	**5 Convs**	**6 Convs**	**7 Convs**	**8 Convs**
1 Conv	0.739%	0.655%	0.638%	0.682%	0.646%	0.766%
2 Convs	0.657%	0.633%	0.554%	0.569%	0.579%	0.560%
3 Convs	0.725%	0.581%	0.656%	0.627%	**0.533%**	0.571%
4 Convs	0.672%	0.556%	0.644%	0.545%	0.671%	0.603%

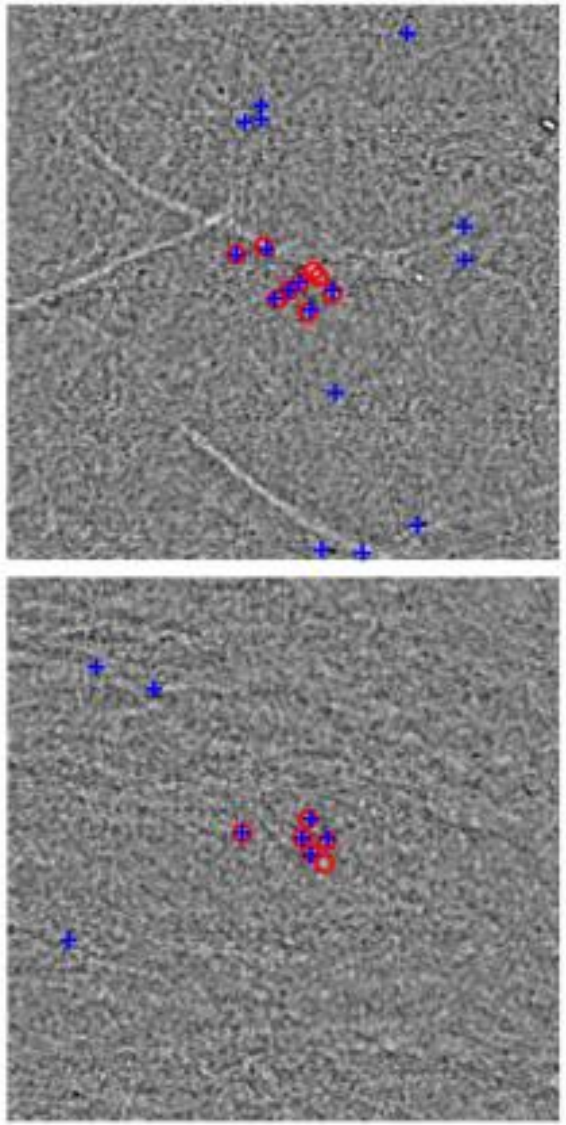
cluster



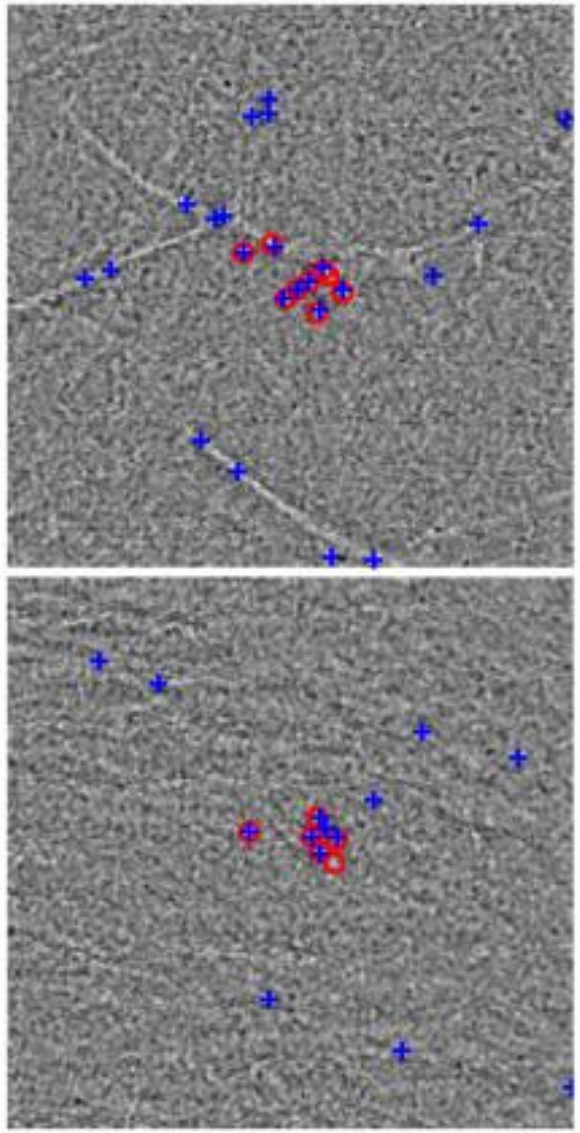
Graphic illustration of Results



(a) Context-sensitive DNN



(b) Unified SVM



(c) Local DNN

Thank you for your attention !